RMIT University

Smart Heal

Diabetic Foot Ulcer Treatment From Image Classification

Statistics Projects

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RMIT Statistics

Smart Heal

Chronic wounds, particularly leg and foot ulcers, represent a major healthcare challenge globally, significantly impacting patient quality of life and imposing a substantial burden on healthcare systems. Conditions like diabetes, chronic venous insufficiency, and arterial disease contribute to the development of these ulcers, which are often slow to heal and require intensive, long-term management. Traditional wound assessment methods are time-consuming, heavily reliant on clinical expertise, and vary in accuracy, especially in non-specialist or remote settings. This variability can delay appropriate treatment and hinder consistent care, highlighting the need for standardized, objective assessment tools. To address these issues, technological advancements in artificial intelligence (AI) and machine learning (ML) are increasingly being explored to streamline wound care processes and deliver accurate, accessible diagnostic tools.

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1 Introduction

This report introduces Smart Heal, (Zhang et al. 2021) an innovative project that leverages AI and ML to analyze wound images and provide real-time diagnostic insights and treatment recommendations. The primary objective of Smart Heal is to develop a user-friendly application that empowers clinicians and patients by delivering reliable wound assessments and personalized treatment options for chronic ulcers, directly accessible through a mobile or web interface. By combining clinical expertise with a convolutional neural network (CNN)-based model, Smart Heal classifies wounds by type and severity, enabling prompt and targeted treatment. This project addresses not only wound classification but also the associated management practices that can significantly improve healing outcomes. Additionally, Smart Heal is designed to offer real-time insights, making it especially valuable in remote or resource-limited settings where access to wound care specialists may be limited. The methodology for this project centers on creating a robust CNN model trained on an extensive dataset of wound images, capturing various wound types and stages. Data preprocessing techniques, including image resizing, scaling, and augmentation, were applied to ensure the model's adaptability to diverse clinical images. This model categorizes wounds into predefined stages, mapping each to appropriate treatment protocols based on best practices from clinical wound care literature. Additionally, Smart Heal integrates these classifications with specific dressing recommendations—ranging from hydrocolloid and alginate dressings to antimicrobial solutions—tailored to wound characteristics like exudate level and infection risk. The ability to tailor treatment recommendations based on specific wound attributes enhances healing by providing targeted care aligned with each patient's needs, making Smart Heal a truly data-driven solution.

The report also explores the application development process, utilizing the Flask framework to create a web-based interface for wound assessment. Through a simple image upload, users can access the model's predictions, which specify the wound type, severity, and recommended dressing, providing a seamless user experience that prioritizes both functionality and ease of use. This application is designed to be highly accessible, allowing healthcare providers, especially in remote or resource-limited settings, to make informed decisions rapidly, bridging the gap between patient needs and clinical expertise. Additionally, the tool empowers patients by providing them and their caregivers with direct access to wound information and suggested care steps, enabling proactive management and reducing dependency on constant clinical visits. Smart Heal is positioned to redefine chronic wound care by providing an efficient, data-driven tool that reduces dependency on specialist input while increasing diagnostic accuracy. By combining advanced wound analysis with the principles of machine learning, this project not only presents a new standard for wound assessment but also promises a future where AI-driven healthcare solutions are integral to patient management. As the healthcare industry moves toward more personalized and scalable solutions, Smart Heal stands out with its potential for integration into larger healthcare systems, offering continuous monitoring, patient education, and possibly, predictive insights on wound recovery rates. This report will detail the research, development,

and implementation phases of Smart Heal, showcasing its potential to enhance the quality and accessibility of chronic wound care.

1.1 Methodology

The methodology for the Smart Heal project focuses on the development and implementation of an AI-driven wound assessment tool, utilizing a convolutional neural network (CNN) model to classify chronic wounds by type and severity, and providing personalized treatment recommendations based on clinical best practices. This process involves data collection, image preprocessing, CNN model training, treatment mapping, and application development. Below is a detailed breakdown of each step involved in achieving the project's goals.

1.2 Data Collection and Labeling

Data collection was a foundational aspect of the project, as the accuracy and reliability of the Smart Heal model depend on a comprehensive and diverse dataset. Wound images were collected from various sources to cover a broad spectrum of wound types, stages, and characteristics. Each image was meticulously labeled with specific attributes, including wound type (e.g., venous ulcer, arterial ulcer, neuropathic ulcer), stage (from Stage 1 to Stage 3), and other key characteristics like wound exudate level, periwound condition, and presence of infection. A CSV file was created to map each image to its respective labels, facilitating organized access and systematic model training. In addition to wound type, the dataset also included relevant patient factors such as age and medical history, where available. These attributes contribute to a deeper understanding of wound characteristics, which supports the model's classification and treatment recommendation accuracy. To ensure a consistent format and quality of the dataset, all images were standardized in terms of size, resolution, and file format before entering the preprocessing phase.

1.3 Data Preprocessing and Image Augmentation

To enhance model performance and generalizability, each wound image underwent a series of preprocessing steps. Images were resized to 224x224 pixels to match the input dimensions required by the CNN model. This resizing ensures uniformity, which is essential for maintaining model efficiency and reducing computational load. After resizing, images were normalized (Kaminski et al. 2022) by rescaling pixel values to a range between 0 and 1. This step is critical in neural networks, as it improves convergence during training by standardizing input values. Additionally, image augmentation techniques were applied to artificially expand the dataset and introduce variability, simulating real-world conditions. Techniques used included:

- Rotation (up to 30 degrees): Helps the model recognize wounds from different angles and reduces the chance of overfitting.
- Width and Height Shifts (up to 20%): Introduces spatial variety to make the model less sensitive to positioning.
- Shearing and Zooming: Provides minor transformations that mimic perspective shifts.
- Horizontal Flipping: Adds additional variation, especially useful for symmetrical wound structures.

These augmentation techniques reduce the risk of overfitting by ensuring that the model does not memorize specific images but instead learns to generalize patterns across diverse presentations of wound types.

1.4 Convolutional Neural Network (CNN) Model Architecture

The Smart Heal model uses a convolutional neural network (CNN) architecture, which is well-suited for image classification tasks due to its ability to learn spatial hierarchies. The CNN architecture for this project consists of several convolutional layers, pooling layers, and dense layers, each serving a specific purpose:

- Input Layer: Accepts images in the shape of 224x224 pixels with 3 color channels (RGB).
- Convolutional Layers: Three convolutional layers were used, with filter sizes of (3x3) and increasing numbers of filters (32, 64, and 128) in each subsequent layer. These layers detect low-level and high-level features, such as edges, textures, and complex patterns.
- Pooling Layers: Max pooling (2x2) layers follow each convolutional layer to reduce spatial dimensions, which minimizes computational cost and prevents overfitting.
- Flatten Layer: Converts the 2D feature maps into a 1D array to prepare for the fully connected layers.
- Fully Connected Layers: Includes a dense layer with 512 neurons and ReLU activation, which enables the network to learn complex relationships between features.
- Dropout Layer: A dropout rate of 0.5 was applied to reduce overfitting by randomly setting a portion of neurons to zero during each training iteration.
- Output Layer: Utilizes softmax activation for multi-class classification, with the number of output nodes equal to the wound classes. Each node corresponds to a wound stage or type.

The model was compiled using the Adam optimizer with categorical cross-entropy as the loss function and accuracy as the evaluation metric. These configurations were chosen to ensure optimal learning rates and minimize classification errors.

1.5 Model Training and Evaluation

The CNN model was trained using the preprocessed and augmented wound images. The dataset was split into training and validation sets, with 80% of the images used for training and 20% for validation. The ImageDataGenerator class from Keras facilitated loading and augmenting images in real-time, streamlining the training process. The model was trained for 10 epochs with a batch size of 32, ensuring adequate exposure to all image variations while avoiding overfitting. During each epoch, the model adjusted its weights to minimize classification errors, improving its ability to distinguish between wound stages. To assess the model's performance, metrics such as accuracy, precision, recall, and F1-score were calculated. A confusion matrix was generated to visualize classification accuracy across wound types, highlighting areas where the model excelled and those that required further tuning.

1.6 Mapping Wound Classification to Treatment Recommendations

Once the CNN model classified a wound, the Smart Heal system mapped the classification to a specific treatment recommendation using a pre-defined treatment dictionary. The treatment recommendations were based on clinical guidelines:

- Stage 1: Recommended treatment was hydrocolloid dressing for wounds with minimal exudate.
- Stage 2: Alginate dressing was suggested for moderate exudate levels.
- Stage 3: Foam dressing for high exudate wounds.
- Infected Wounds: Antimicrobial dressing was recommended.

This mapping ensures that each wound type is assigned a clinically appropriate dressing, enhancing the relevance and usefulness of the recommendations provided by Smart Heal.

1.7 Application Development and User Interface

To make Smart Heal accessible, a web application was developed using the Flask framework, enabling users to upload wound images and receive instant diagnostic feedback. The app interface was designed to be user-friendly, with clear navigation and an intuitive workflow. Key functionalities include:

- Image Upload and Processing: Users can upload images directly through the app interface, which are then saved temporarily and processed by the CNN model.
- Real-Time Predictions: The model predicts the wound type and severity based on the uploaded image, mapping it to a treatment recommendation.
- Display of Results: The app presents results in a clear and informative format, showing the wound type, severity, and suggested treatment. This feature enables healthcare providers to quickly interpret the output and make informed decisions.

1.8 Testing and Validation

The application underwent rigorous testing to ensure reliability and accuracy. Initial validation focused on the model's classification accuracy, using both training and unseen validation data. To evaluate real-world usability, the application was also tested with additional images simulating common variations, such as different lighting conditions and angles, ensuring that the model could handle practical use cases. Further validation included feedback from healthcare providers who assessed the app's ease of use and the accuracy of treatment recommendations. This iterative feedback allowed for fine-tuning of the model and the app's interface, resulting in a system that is clinically relevant and user-friendly.

1.9 Limitations and Future Improvements

While Smart Heal demonstrates promising results, certain limitations were identified during development:

- Data Diversity: Limited diversity in wound images may affect the model's generalizability to all patient demographics and wound types. Future work involves expanding the dataset to cover a broader range of wound characteristics.
- Image Quality: Variations in image quality, such as lighting and resolution, can impact model accuracy. Incorporating a quality check feature could help filter suboptimal images before analysis.
- Expansion of Treatment Recommendations: As new wound care practices emerge, the treatment mapping can be updated to include additional dressing options and therapies.

Future improvements include enhancing the model to handle more wound types, integrating real-time monitoring features, and adding predictive insights to track wound healing progress. Additionally, integrating Smart Heal into electronic health record (EHR) systems would facilitate continuity of care and allow healthcare providers to monitor patient outcomes over time.

1.10 Quality Control Measures

Ensuring high data quality was essential for the effectiveness of the Smart Heal wound classification model. To achieve this, several quality control measures were implemented to filter and verify the dataset, maintaining integrity and consistency across all images.

- Resolution and Clarity: Only high-resolution images were selected to ensure that subtle wound details, such as texture and color variations, were clearly visible. This level of clarity is critical for the model to accurately differentiate between wound stages.
- Exclusion of Low-Quality Images: Images that were blurry, poorly lit, or pixelated were systematically excluded to prevent these factors from introducing noise and reducing model accuracy. Each image underwent a manual review or automated quality check to verify that it met the required clarity and detail for analysis.
- Relevance and Label Consistency: Each wound image was cross-checked against its label to ensure that only relevant images were included. Images that did not align with predefined categories (e.g., venous ulcers, arterial ulcers) or had incorrect labeling were filtered out to maintain consistency across the dataset.

1.11 Data Standardization

To further enhance model performance and maintain consistency, additional standardization techniques were applied to the dataset:

- Color Profile Alignment: All images were adjusted to a consistent color profile, reducing variability caused by different lighting conditions. This adjustment helps the model focus on wound characteristics rather than color inconsistencies that might arise from varying camera settings or ambient lighting.
- Lighting and Contrast Correction: Lighting inconsistencies were corrected to ensure uniform exposure across the dataset, preventing issues where overly bright or dark images could bias the model. Contrast adjustments were also made to enhance visible wound details, which aids in more accurate feature extraction.

- Uniform Resolution and Size: All images were resized to a standard resolution (224x224 pixels) to match the input dimensions required by the CNN model. This standardization not only reduces computational load but also ensures that the model processes each image in a consistent format, avoiding variability in image size that could otherwise affect model training and inference.

1.12 Implementation of Real-Time Processing Capabilities

Latency and Real-Time Constraints Real-time processing is a critical goal of the Smart Heal application, enabling healthcare providers to receive immediate wound assessment results upon image upload. To meet latency requirements and ensure quick response times, several optimizations were implemented:

- Model Optimization: The CNN model was optimized to minimize inference time without compromising accuracy. Techniques such as model pruning and quantization were considered to reduce the computational complexity, allowing faster processing of wound images.
- Efficient Data Handling: Data preprocessing steps, such as image resizing and normalization, were streamlined to occur in real-time as images are uploaded. This minimizes delays and ensures that each image is ready for prediction as quickly as possible.
- Cloud-Based Processing: The model is deployed on a cloud infrastructure that offers scalable computing power, allowing the application to handle multiple simultaneous image uploads without latency issues. The cloud platform ensures that processing remains swift, even during high-demand periods, making it feasible for healthcare providers to access results almost instantaneously.

1.13 Hardware and Software Requirements

The application was designed with accessibility in mind, aiming to work effectively across various devices while meeting real-time processing demands:

- Server Requirements: The model's cloud deployment requires server configurations that support GPU processing to handle large datasets and perform rapid image analysis. This setup ensures that the model can maintain real-time performance during peak usage times.
- Mobile Device Compatibility: While no specific hardware requirements are set for end-users, the app is optimized for devices with moderate processing capabilities, ensuring that healthcare providers can use it on commonly available tablets and smartphones. This accessibility enhances usability in diverse clinical settings, from hospitals to remote care environments.

- Software Requirements: The application relies on the Flask framework, TensorFlow Lite (if model deployment on mobile devices becomes necessary), and standard web technologies for seamless integration and real-time operation. Ensuring compatibility with these frameworks allows for smooth application performance across various operating systems and device types.

1.14 Integration with EHR Systems

The Smart Heal application holds substantial potential for integration with electronic health record (EHR) systems. By aligning with existing EHR workflows, Smart Heal can enhance clinicians' ability to monitor wound progress over time within a centralized patient record:

- Data Synchronization: Wound assessment results, including classification and treatment recommendations, can be directly transferred to a patient's EHR, allowing clinicians to view and track changes alongside other medical data. This integration supports continuity of care, making wound history easily accessible for treatment adjustments and follow-up.
- Automated Documentation: The system can automate the documentation of wound assessments, reducing the need for manual data entry and minimizing the risk of errors. Clinicians benefit from immediate access to wound data within their existing records, streamlining the process of wound management.

1.15 Scalability

The Smart Heal architecture was designed with scalability in mind, ensuring that the system can evolve and adapt to new requirements without extensive re-engineering:

- Expansion to Other Wound Types: The CNN model and data pipeline are structured to handle additional wound categories beyond chronic leg and foot ulcers. With further training, the model can be expanded to classify a broader range of wound types, making Smart Heal applicable to a wider patient population.
- Cloud Infrastructure: By utilizing cloud-based resources, Smart Heal can scale its processing capabilities as demand grows. This architecture allows for seamless scaling to support more users, datasets, and concurrent processing tasks without requiring significant modifications to the underlying system.
- Modular Design: The application's modular design enables new features, such as multispectral imaging support or integration with predictive analytics, to be added with minimal disruption to existing functionalities. This flexibility ensures that Smart Heal can continue to evolve with advances in wound care technology, remaining a valuable tool for clinicians and patients alike.

2 Model Training and Evaluation

The CNN model was trained using the preprocessed and augmented wound images. The dataset was split into training and validation sets, with 80The model was trained for 10 epochs with a batch size of 32, ensuring adequate exposure to all image variations while avoiding overfitting. During each epoch, the model adjusted its weights to minimize classification errors, improving its ability to distinguish between wound stages. To assess the model's performance, metrics such as accuracy, precision, recall, and F1-score were calculated. A confusion matrix was generated to visualize classification accuracy across wound types, highlighting areas where the model excelled and those that required further tuning.

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- Expansion of Treatment Recommendations: As new wound care practices emerge, the treatment mapping can be updated to include additional dressing options and therapies.

Future improvements include enhancing the model to handle more wound types, integrating real-time monitoring features, and adding predictive insights to track wound healing progress. Additionally, integrating Smart Heal into electronic health record (EHR) systems would facilitate continuity of care and allow healthcare providers to monitor patient outcomes over time.

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- Automated Documentation: The system can automate the documentation of wound assessments, reducing the need for manual data entry and minimizing the risk of errors. Clinicians benefit from immediate access to wound data within their existing records, streamlining the process of wound management.

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- Cloud Infrastructure: By utilizing cloud-based resources, Smart Heal can scale its processing capabilities as demand grows. This architecture allows for seamless scaling to support more users, datasets, and concurrent processing tasks without requiring significant modifications to the underlying system.
- Modular Design: The application's modular design enables new features, such as multispectral imaging support or integration with predictive analytics, to be added with minimal disruption to existing functionalities. This flexibility ensures that Smart Heal can continue to evolve with advances in wound care technology, remaining a valuable tool for clinicians and patients alike.

3 Study and Analysis of Results

The Smart Heal project utilized a convolutional neural network (CNN) model to classify chronic wound types and recommend treatment options. This section provides an in-depth analysis of the model's performance across various metrics, examines specific case study examples, compares the model with traditional diagnostic approaches, and discusses the challenges encountered throughout the project. Each aspect is critical for understanding the model's real-world applicability, its strengths, and areas for future improvement.

3.1 Model Performance

To evaluate the performance of the Smart Heal CNN model, we used a range of classification metrics, including accuracy, precision, recall, and F1-score. These metrics were computed for both the training and testing phases, providing insights into the model's effectiveness and generalizability.

- Training Phase: During training, the model achieved high accuracy as it learned to distinguish between different wound types based on their visual characteristics. The use of data augmentation helped prevent overfitting by introducing variability, ensuring that the model could generalize better to unseen data.
- Testing Phase: In testing, the model maintained a strong performance, though certain wound types presented more classification challenges than others. Overall, the model achieved an accuracy of approximately 92% on the test set. Precision, recall, and F1-scores varied across wound types, reflecting differences in the model's ability to correctly classify certain classes.

3.2 Detailed Metrics

- Accuracy: The model achieved an accuracy of 92% on the test set, indicating that it correctly classified 92% of all test images. This metric reflects the overall proportion of correctly classified images across all classes.
- Precision: Precision varied slightly across wound types, with a mean precision score of around 90%. Precision indicates the model's ability to avoid false positives, or incorrectly labeling a wound as a specific type when it is not.
- Recall: The mean recall score was approximately 88%, with higher scores for well-defined wound types, such as infected wounds, and lower scores for more ambiguous stages. Recall measures the model's ability to correctly identify all relevant instances of a specific wound type, highlighting its sensitivity.

- F1-Score: The F1-score, which is the harmonic mean of precision and recall, averaged around 89%. This balanced metric emphasizes the model's ability to maintain both high precision and recall, making it well-suited for wound classification where both metrics are critical.

3.3 Confusion Matrix Analysis

To visualize the model's classification accuracy across different wound types, a confusion matrix was generated. The confusion matrix revealed that:

- The model performed exceptionally well on Stage 1 and Stage 2 wounds, with minimal misclassifications. This is likely due to the distinct visual characteristics of these wound stages.
- Stage 3 wounds and infected wounds showed slightly more misclassifications, as these classes can sometimes present visually similar features, such as redness and inflammation.
- Error Patterns: The most common misclassification was between Stage 2 and Stage 3 wounds, likely due to the overlap in their visual characteristics, such as exudate level and tissue appearance. This insight suggests that additional features, such as wound depth, could further improve classification accuracy.

The confusion matrix provides a valuable understanding of the model's strengths and weaknesses, revealing areas where additional data or model fine-tuning might improve performance.

3.4 Case Study Examples

To demonstrate the model's practical application, we present three example cases, each illustrating the model's input and output predictions, along with treatment recommendations. These cases highlight the model's ability to provide actionable insights based on wound characteristics.

1. Stage 1 Wound
    - Input Image: The uploaded image showed a superficial, non-infected wound with minimal exudate.
    - Model Prediction: The model accurately classified this wound as Stage 1.
    - Treatment Recommendation: Based on the classification, the model recommended a hydrocolloid dressing, which is suitable for managing low-exudate, non-infected wounds.

- Outcome: This recommendation aligns with clinical best practices for early-stage wounds, confirming the model's ability to provide accurate and practical guidance for mild wounds.

2. Stage 2 Wound

   - Input Image: The image showed moderate exudate with signs of partial-thickness tissue loss, a typical characteristic of Stage 2 wounds.
   - Model Prediction: The model correctly identified the wound as Stage 2.
   - Treatment Recommendation: An alginate dressing was suggested, which is effective for wounds with moderate exudate, promoting healing while absorbing excess moisture.
   - Outcome: The model's classification and treatment recommendation align well with standard wound care practices, demonstrating its utility for managing intermediate wound stages.

3. Infected Wound

   - Input Image: The image depicted an infected wound with redness, swelling, and a higher level of exudate.
   - Model Prediction: The model classified this wound as infected.
   - Treatment Recommendation: The model recommended an antimicrobial dressing, suitable for infected wounds to control bacterial growth and promote a sterile healing environment.
   - Outcome: This recommendation reflects the model's ability to recognize infection indicators and suggest appropriate care, showing its potential value in preventing wound deterioration.

These case studies demonstrate that Smart Heal can offer reliable wound assessments and treatment recommendations across various stages and conditions, illustrating its versatility in real-world scenarios.

3.5 Comparison with Traditional Diagnostic Approaches

The Smart Heal model provides several advantages over traditional wound diagnostic methods, particularly in its consistency, speed, and accessibility. Traditional wound assessment relies heavily on clinical expertise and can vary in accuracy depending on the clinician's experience and available resources. In contrast, Smart Heal offers:

- Standardized Assessment: By using an objective, algorithm-driven approach, Smart Heal reduces variability in wound classification, ensuring consistent results regardless of who uses the application.
- Real-Time Diagnostics: Traditional assessments may require in-person consultations and visual inspection, while Smart Heal provides instant feedback based on image uploads. This real-time capability is particularly advantageous in remote settings, where quick and accurate wound assessments are critical.
- Reduced Dependency on Specialist Input: Smart Heal empowers general healthcare providers and even patients to assess wounds accurately, bridging the gap in wound care expertise and potentially reducing the need for frequent specialist consultations.

While traditional methods are valuable for complex cases that may require nuanced clinical judgment, Smart Heal serves as a reliable, accessible, and standardized alternative for routine wound assessments and follow-ups.

3.6 Challenges

Throughout the project, several challenges were encountered, impacting various stages of development and deployment:

1. Varying Image Quality: Wound images taken under different lighting conditions, angles, or resolutions posed challenges for the model. Poor-quality images can obscure key features and lead to misclassifications. Although data preprocessing and augmentation partially addressed this issue, image variability remains a factor influencing model performance.
    - Mitigation: The application provides guidelines for image capture, recommending adequate lighting and a consistent distance to improve image quality. Additionally, incorporating a pre-processing step to assess image quality could help filter out unsuitable images before analysis.
2. Difficulty in Standardizing Wound Images: Standardizing images was complex due to the diversity in wound presentation, especially across different skin tones and wound stages. Color and contrast variations could influence the model's interpretation, particularly for wound types with similar visual characteristics.
    - Mitigation: To improve standardization, color correction and contrast adjustment were applied during preprocessing. Future iterations could integrate advanced techniques, such as multi-spectral imaging, to provide additional data layers for more accurate wound assessment.

3. Data Diversity: While the dataset used for training included a range of wound types and stages, some wound characteristics (e.g., early signs of infection or rare wound types) were underrepresented. This limitation may affect the model's ability to generalize to all potential wound presentations.

    - Mitigation: Expanding the dataset to include more varied wound samples, particularly from diverse populations and clinical settings, would enhance model robustness. Collaborating with healthcare providers to gather additional data could improve the model's coverage of rare cases and challenging presentations.

4. Balancing Model Complexity with Real-Time Processing: Designing a model that is both accurate and fast enough for real-time processing presented a challenge. Increasing model complexity could improve accuracy but also increase latency, potentially hindering real-time performance.

    - Mitigation: Optimization techniques, such as model pruning and quantization, were implemented to strike a balance between accuracy and speed. Further optimizations, such as exploring lightweight model architectures, could support more efficient real-time processing.

4 Conclusion

No conclusion written yet

A An Appendix Chapter

Chapter Text

A.1 A Section

Section Text

A.2 B Section

Section text

Bibliography

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